

SUMMARY

AI Engineer and PhD researcher building privacy-preserving ML and generative AI systems with Python, PyTorch, FastAPI, Docker, and MLOps. Focused on synthetic data, differential privacy, LLM applications, and production-ready evaluation pipelines.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
GenAI & LLMs: RAG, Vector DBs, Prompt Engineering
Databases: MySQL, PostgreSQL, SQLite, MongoDB
DevOps & Tools: Docker, FastAPI, REST APIs, Git, CI/CD

PROFESSIONAL EXPERIENCE

Research Assistant Sep. 2025 – Present
Eötvös Loránd University (ELTE) Budapest, Hungary

- Build and optimize DP-SGD federated-learning pipelines for privacy-preserving synthetic data generation ($\epsilon = 1$), including data preparation, feature engineering, and privacy-utility evaluation
- Author and maintain the open-source IDaSec Practicum: 13 hands-on labs across 8 modules covering ML attacks, defenses, privacy, and security auditing
- Teach secure ML and mentor graduate students in AI engineering, reproducible experimentation, and MLOps workflows

AI Engineer Dec. 2025 – Present
MyVellum  Remote

- Built and containerized a FastAPI RAG service with Groq-hosted LLMs, vector-backed retrieval, and modular prompt orchestration
- Implemented CI/CD evaluation gates that block deployments when retrieval-quality metrics fall below defined thresholds

Data Analyst Dec. 2024 – Mar. 2025
Djezzy Khenchela, Algeria

- Analyzed subscriber telecom datasets using Python and SQL to identify usage trends and customer behavior patterns
- Built automated data preparation and feature engineering pipelines, facilitating data-driven decision making and reporting
- Collaborated with analytics and cross-functional teams to model subscriber metrics and optimize customer retention strategies

PROJECTS

CASPER  2026
Python, Differential Privacy, RAG, T5, MedCPT, Docker

- Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
- Evaluated the privacy-utility trade-off, achieving Recall@10 = 0.91 and MIA AUC $\approx$ 0.50 (random-guessing)

Context-Aware Access Control for IoT Devices  2026
FastAPI, Python, Multi-Agent LLMs, Docker, Wazuh, Prometheus ITAT'2026, WAS

- Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter (< 1 ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
- Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts

EDUCATION

PhD in Informatics (GPA: 5/5) Sep. 2025 – Expected 2029
Eötvös Loránd University (ELTE) Budapest, Hungary

Thesis: Applied synthetic data generation for secure, privacy-preserving AI systems (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence – Ranked 1st in Class (GPA: 3.7/4.0) Sep. 2023 – Jun. 2025
University of Khenchela Khenchela, Algeria

Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)

B.Sc. in Mathematics and Computer Science – Ranked 1st in Class (GPA: 3.9/4.0) Sep. 2020 – Jun. 2023
University of Khenchela Khenchela, Algeria

Final Year Project: A learning management system for Algerian students. 

AWARDS & CERTIFICATIONS

- **Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE Sep. 2025
- **2nd Place — Starthon'25 Hackathon:** University of El Oued, Algeria Apr. 2025
- **Software Engineering (Back-end Specialization):** ALX Africa  Feb. 2025